

18.06 Problem Set 10

Due Friday, November 18 at 11am.

Problem 1 (5+5+5+5 points)

In class, we saw that $\mathbf{o} = [1, 1, \dots, 1, 1]$ is an eigenvector of M^T with eigenvalue $\lambda = 1$ for any Markov matrix M .

(a) If x_k is an eigenvector of M ($Mx_k = \lambda_k x_k$) for any *other* eigenvalue $\lambda_k \neq 1$ of M , show that we must have $\mathbf{o}^T x_k = 0$: it must be *orthogonal* to $\mathbf{o}$. (Hint: use $\mathbf{o}^T = \mathbf{o}^T M$.)

(b) Check your result from (a) numerically for a random 5×5 Markov matrix `M = rand(5,5); M = M ./ sum(M, dims=1)`, with eigenvalues `eigvals(M)` and eigenvectors `X = eigvecs(M)`. (Do `using LinearAlgebra` to get `eigvecs` and `eigvals`.)

(Note: if you have a long vector `v`, Julia only shows a few elements by default, but you can show all the elements with `@show v`. You can also look at the absolute values of the elements with `abs.(v)`, which can be easier to read than complex numbers in checking that entries are small.)

(c) If we expand an arbitrary x in an eigenvector basis $x = c_1 x_1 + \dots + c_m x_m$, letting x_m be a steady-state eigenvector ($\lambda_m = 1$) and supposing all of the other eigenvalues are $\neq 1$, show that $\mathbf{o}^T x$ gives us a simple formula for $c_m = \underline{\hspace{2cm}}$.

(d) Hence, if all other eigenvalues have magnitude < 1 , then $M^n x \rightarrow \underline{\hspace{2cm}}$ (simple formula in $\mathbf{o}, x, x_m$) as $n \rightarrow \infty$. Check this formula against `M^100 * [1,2,3,4,5]` for your `M` from (b).

Problem 2 (10+5 points)

From Strang, section 6.2. Consider $A = \begin{pmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$ and $B = \begin{pmatrix} 0.6 & 0.9 \\ 0.1 & 0.6 \end{pmatrix}$. For this problem you keep in mind the diagonalization of matrices like A and B .

(a) Which of A^n or B^n (or both, or neither) go $\rightarrow \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ as $n \rightarrow \infty$? Double-check your answer by looking at A^{100} and B^{100} in Julia — these are approaching matrices of rank $\underline{\hspace{2cm}}$ and $\underline{\hspace{2cm}}$, respectively.

(b) For what values of the real scalar μ is $\sqrt{A - \mu I}$ a real matrix? Check your answer by trying `sqrt(A - μ*I)` for a few values of `μ` in Julia (do `using LinearAlgebra` to get `I`).